Conditional generation of cathode vacancy patterns at requested compositions and energy ranges

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Abstract

We study whether conditional variational autoencoders (CVAEs) efficiently propose new Li/vacancy arrangements at specified compositions and energy ranges in a fixed layered $LiCoO_2$ parent. Exact-count decoding enforces composition, while energy control must be learned and independently measured. Earlier binary and multibin models enriched low energies relative to uniform sampling but lost to frozen low-energy-seed mutation. The extension separates training-policy, latent-size, symmetry, and physical-neighbor effects, and audits whether deletion energies represent relaxed ordering. Causal decoder neighborhoods improve final requested-bin yield over frozen mutation under UMA and MACE, while latent-use diagnostics limit attribution to the variational pathway. Teacher-defined success is distinguished from DFT accuracy and relaxed geometry generation.

1 Physical motivation, revised question, and data

Direct cathode recycling seeks to restore useful composition and structure; hydrothermal treatment followed by annealing can regenerate degraded $LiCoO_2$ (LCO) [6]. Our longer-term application is to select informative structures for reference calculations and better lithium-replenishment models. The class experiment studies conditional generation, rather than simulating the recycling liquid or reactor. Li/vacancy ordering changes energies at a fixed Li content [8], creating a large discrete search space.

Density functional theory (DFT) supplies an approximate electronic-structure reference at specified species, coordinates, and cell. A machine-learned interatomic potential (MLIP) approximates the energy and its force/cell derivatives. A single-point call keeps coordinates fixed; relaxation moves them toward a local minimum. Molecular dynamics (MD) uses forces to propagate atoms. Drawing a vacancy pattern performs none of these physical operations.

The parent is $Li_{48}Co_{48}O_{96}$; retaining n Li gives $N_a = 144 + n$ present atoms and fraction $x = n/48$. The generated object is an occupation vector $o \in \{0,1\}^{48}$, with $o_j = 1$ for Li present and vacancy $v_j = 1 - o_j$. Li indices are full-parent rows 48–95, zero based. Primary counts are $n = 12, 24, 36$. The main labels are energies of pristine-derived deletions, not relaxed minima. UMA-S-1.2.1/`omat`, FP32 with TF32 off, defines the primary new-label teacher; matched MACE checks use a separate energy namespace.

1.1 Independent examples and protected roles

The parent's 192 retained permutations define an orbit: all related occupation patterns under this fixed host's symmetries. Its canonical key is the smallest transformed vacancy string. Grouping precedes splitting; augmentation does not create independent examples. At half filling, 32,247,603,683,100 raw masks give 168,025,313,938 orbits for this cell/group, not for every supercell shape. The frozen round-0 data have 40,157 training, 4,967 validation, and 5,123 test orbits. New development evidence selects policy; a fresh final protocol/history is frozen before proposals. Previously inspected assessments cannot be renamed untouched tests. No new DFT or MLIP fine-tuning campaign is run.

A Pt–Ni study already combines occupation-based CVAEs, property classes, and UMA evaluation [9]. Our contribution is an interpretable, symmetry-aware test against strong composition-preserving mutation, including negative results. The software interface describes a general parent/mobile-ion sublattice, but present physical evidence is LCO-specific.

2 Conditional model, exact count, and physical graph

At each composition, training-only energies $e_i = E_i/N_a$ (eV/atom) define nearest- rank q_{20} and q_{80} boundaries. Low is $e \le q_{20}$, middle is $q_{20} < e \le q_{80}$, and high is $e > q_{80}$. Ties stay together; unsupported joint requests fail explicitly. Each schema records parent, teacher/head, precision, units, geometry protocol, boundaries, support, and input hashes.

Model family	Supplied information and distinction
Historical composition CVAE	$c = [x]$; no energy request.
Historical binary energy CVAE	$c = [x, \mathbf{1}_{\text{low}}]$; middle/high both mean non-low.
Three-bin CVAE	$c = [x, \text{onehot}(b)]$; distinct low/middle/high requests.
No-latent AR control	Same sequential/count mechanism without variational latent variables.
Physical-neighbor variants	Distinguish learned graph encoder from causal graph-derived decoder summaries; retain the same count constraint.
Seed-edit CVAE, if executed	Additional seed and specified swap count; local editing rather than de novo generation.

The encoder $q_\phi(z \mid o, c)$ predicts a Gaussian mean μ and diagonal variance σ^2. A GRU decoder of hidden size 64 defines

$$p_\theta(o \mid z, c) = \prod_{j=1}^{48} p_\theta(o_j \mid o_{<j}, z, c), \qquad z \sim \mathcal{N}(0, I) \text{ at generation.} \tag{1}$$

The decoder receives conditions and latent context at every step. During fitting, previous true bits provide teacher forcing; during generation they are sampled. If the remaining count forces a bit, its conditional probability is one and its reconstruction penalty zero. Energy membership has no such hard guarantee.

The fitted objective is

$$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}[-\log p_\theta(o \mid z, c)] + \frac{\beta}{2} \sum_{k=1}^{D} \left(\mu_k^2 + \sigma_k^2 - \log \sigma_k^2 - 1\right). \tag{2}$$

The first term reconstructs masks; the second is KL divergence to the sampling prior. D is latent dimension and β its dimensionless regularization weight. Training β warms to 0.1; alternative checkpoint criteria use validation weights 1 and 0.1 on the *same trajectory*. The KL sum is not silently divided by D. Neither good reconstruction nor large KL establishes useful prior samples [5, 7, 1]. Class balancing changes training-only row weights while preserving their total; the same weights average reconstruction and summed KL. Validation retains base weights and fixed two-draw latent noise, independent of training randomness.

2.1 What spatial feature extraction changes

A reusable parent schema stores host species, all possible mobile sites, positions/cell in Å, periodic directions, valid symmetries, and graph edges with integer image shifts. Vacant sites remain nodes. Distances come from the crystal, not adjacent sequence indices. Message passing learns combinations of neighbor features and pools them into latent parameters [3]. Backpropagation differentiates the CVAE loss through these operations; it does not minimize a physical energy. An encoder-only graph is absent during prior generation. Decoder neighborhood summaries therefore use only already assigned occupancies and an explicit unassigned state. Future target bits or full-mask encoder embeddings must not bypass the latent channel. An indexed GRU with graph features is not automatically permutation/rotation equivariant. The context comparison adds decoder summaries to the same graph-encoder control; an initial fit was replicated at two more seeds before the fresh final assessment.

3 Experiments, observables, and scientific controls

The full protocol, commands, selected configurations, and per-experiment status are supplied in the portable packet. Historical and new assessments retain different scope labels. The new design first compares class balancing off/on and both checkpoint rules at a fixed update budget with three fit seeds. Matched latent dimensions 4, 8, 16, 32 reuse the selected dimension 8 trajectory; a no-latent AR control isolates the latent pathway. A canonical-versus-training-augmentation comparison precedes graph comparisons, so orientation and graph effects are not silently combined. Parameter counts and actual updates are saved.

Latent diagnostics include summed/per-dimension KL, encoded-mean variance across configurations, and prior-sample usefulness. "Active" components use declared mean-variance cutoffs with sensitivity reported. These are descriptive diagnostics, not a theorem of collapse or a physical convergence test [2, 4]. The latter paper analyzes a simplified linear VAE, not this crystal network. Posterior reconstructions and prior generation answer different questions. Paired low/middle/high requests share latent vectors and AR random streams; separate changes of z test latent sensitivity.

3.1 Baselines and complete accounting

Uniform sampling is uniform over raw fixed-count masks. Frozen mutation selects a protected training seed uniformly in the requested bin, selects one/two/three swaps uniformly, then selects distinct occupied removals and vacant insertions uniformly. It has no distance restriction, energy acceptance, or online population update. Every proposal is frozen before scoring. Method names containing "adapt" can identify the source seed dataset and do not imply adaptive search. An edit model must receive the same seeds and swap distribution as mutation; joint seed/child transformations preserve edit alignment.

For requested bin b, the primary useful yield is

$$Y_b = \frac{\#\{\text{distinct novel orbits measured in requested bin } b\}}{\#\{\text{all attempts requesting } b\}}. \tag{3}$$

Known masks, repeats, invalid outputs, and failures remain attempts. Report novelty to training and to all frozen history separately, confusion by composition and fit seed, energy distributions, distances from training/seeds, and useful energy versus attempts and actual new scoring calls. Fixed-original-q_{20} yield is a separate requested-low metric. Fit seeds are independent model replicates; latent draws are not. Teacher agreement is not independent DFT truth.

3.2 Two audits before interpreting new training

The energy-target audit pairs existing deletion and fixed-cell endpoint energies for the same masks, retaining failures and protected lineages. It tests rank and low-tail changes, Li-site reassignment, and endpoint occupation-class merges. Occupation classes do not certify unique potential-energy basins. A second audit uses a symmetry-invariant structural descriptor: the nearest-shell fraction of eligible mobile-site pairs that are both vacant. Pairs are consistently counted once; vacancy is $1 - o_j$. Train-derived descriptor ranges and joint support are checked before any joint-condition pilot. Mutation receives the same request.

Static temperature acquisition remains secondary. Its exponent uses total-cell energy gaps, with orbit multiplicity for exhaustive representatives or retained raw draw frequencies for a raw-mask reference, never both. Proposal-biased training histograms and latent probabilities do not measure physical entropy. The prior reference has low effective sample size and identical coarse request allocations across temperatures; those limitations are retained.

4 Measured results and interpretation

Development selects a spatial conditional sampler. Six dimension 8 fits tested balancing and both checkpoint rules on shared trajectories; twelve more tested dimensions 4, 16, 32 and no-latent AR controls. Balancing and selection weight 0.1 were retained; increasing latent size improved reconstruction without monotonic discovery benefit. The frozen size rule retained D4 using practical 1 percentage-point balanced and 2 point low-yield margins, not confidence bounds. Graph-context development gave 60.88% balanced yield versus mutation's 58.80%; its initial fit was replicated at two more seeds before final generation. All 28 core fits completed 4000 updates, within the 30-fit cap. Full policy/size tables and individual seeds are supplied separately.

Frozen final assessment. The method list and 278,382-orbit history were frozen before 34,560 UMA attempts. Six prespecified families, the same 20,736 attempts, were also evaluated with MACE:

Frozen-final method	UMA in-bin (%)	MACE in-bin (%)	UMA low (%)
Dense D4	31.05 ± 0.81	31.25 ± 0.99	30.30
Dense + symmetry	33.19 ± 1.10	33.16 ± 1.14	29.08
Graph encoder	34.43 ± 0.18	34.55 ± 0.09	25.09
Graph + causal context	62.04 ± 1.70	63.45 ± 1.51	63.54
Uniform	34.14 ± 0.36	34.03 ± 0.17	19.97
Frozen mutation	58.62 ± 0.51	57.29 ± 0.78	52.52

Rows average counts/bins equally, then three fit or sampler replicates; ± is descriptive seed standard deviation. "In-bin" measures distinct novel orbits in the requested teacher-specific class; "low" uses only low requests and the original UMA q_{20}. MACE remeasures UMA-requested proposals under its own training-derived boundaries, not a separately retrained MACE generator. Its original-q_{20} low yields are 67.62% for context versus 52.95% mutation. UMA balanced gain over mutation is 3.41 percentage points, versus 11.02 points on its low metric. This is a teacher-defined sampling advantage, not DFT accuracy.

Paired low/middle/high requests share z and AR uniforms. For graph context, high-minus-low energy averages +5.263 meV/atom, positive in 97.40% of pairs; middle-minus-low averages +2.100 meV/atom. Dense D4's corresponding means are +0.026 and −0.135 meV/atom. Actual-checkpoint tests found zero future-bit gradients and unchanged logits under future-mask changes, supporting the tested causal implementation rather than an obvious future-label shortcut.

Better conditional sampling is not evidence of useful latent variables. Graph-context validation KL is only 0.007–0.011 nats; no pooled mean component exceeds variance 0.01. Resampling z at fixed uniforms changes 0.795 sites on average and leaves 68.23% in the same orbit. Posterior/prior Hamming distances are 16.78/16.81. In contrast, dense D4 has KL 11.58–11.95 nats and changes 12.72 sites. Thus the positive result primarily supports a condition-sensitive spatial AR sampler; the variational pathway is nearly unused by these diagnostics. An explicit matched no-latent spatial control is a discriminating later test.

Symmetry-minimized Hamming distances to a fixed 1024-training-orbit reference per count average 6.858 sites for context, 6.891 uniform and 6.498 mutation. These are upper bounds on full-training nearest distance, not evidence of a large diversity gain. All 3456 mutation attempts have their exact $2k$ edits verified.

The second structural condition remains a negative pilot. One middle-energy fit requesting low/high vacancy-pair fractions gave 28.13% joint useful yield, versus 28.52% plain CVAE and 44.79% matched mutation. The descriptor strongly correlates with teacher energy; it is not an independent thermodynamic property. No multi-fit improvement is claimed. This separate pilot and the paired conditional diagnostics are development evidence; the table is the frozen final assessment.

5 Energy target, limits, and remaining team contributions

The independent audit checks all 1000 attempted fixed-cell UMA lineages in the existing archive, not just selected successes. All converge under the recorded 0.05 eV/Å maximum-force target; the largest endpoint maximum force is 0.049990. Initial/final scalar energies agree with 1000 executed source records, and 20 used worker receipts agree on checkpoint, head, precision and inference settings. Cells match the parent exactly. No Li or host nearest-parent assignment changes, site collisions, or occupation-orbit merges occur. There remain 1000 endpoint occupation classes; this does not certify 1000 distinct converged PES basins.

n_{Li}	Paired masks	Spearmanρ	Low20 overlap	Median gain(meV/atom)
12	23	0.9022	4/5	−29.91
24	23	0.9723	4/5	−19.86
36	23	0.9140	4/5	−11.03

Across 45 composition pools, median rank correlation is 0.9379, minimum 0.6383, and minimum low-quintile recall 0.40. These are acquisition-selected examples, not unbiased coverage. Relaxation therefore changes some rankings even while parent-site occupation classes remain stable. At each primary count, four of five cohort-low-quintile members are shared; these cohort thresholds describe this audit and are not new training boundaries.

A matched label-target CVAE fit is deferred: the three primary counts contain only 49 training, 11 validation and 9 test lineages. Training-low endpoint classes have 4, 4, 3 examples; validation-low classes have 0, 1, 0. Repeated intermediate frames do not enlarge that independent support. A later test must retain those roles, acquire additional paired development lineages, fit separate training-only deletion/endpoint thresholds, and evaluate both models under both targets. The existing coordinate CVAE remains separate: its median optimizer saving (40 to 24 steps) was also achieved by deterministic regression.

5.1 Physical validation and generalization boundary

The archived matched DFT audit remains informative but cohort dependent: UMA's median rank correlation is 0.97366 over seven historical O3 rattled pools and 0.73177 over ten Hu O3 optimized pools. At $Li_{10}Co_{12}O_{24}$, it selects a structure ranked last by DFT in the complete five-member pool. High correlation on displaced frames does not guarantee ordering of nearly degenerate minima. Historical/Hu DFT parameters differ, and their absolute energies are not pooled. Good ranks do not certify forces, saddle points, or thermal weights.

No result establishes global minima, physical entropy, reactor optima, downstream MLIP improvement, or zero-shot material/size transfer. A generic parent interface and synthetic alternate-parent checks establish applicability of code, not accuracy on new chemistry. A structure can have many Li hops: a barrier label requires an occupied source, empty destination, periodic displacement, and a consistent geometry/teacher protocol. Any later MLIP NEB pilot is a feasibility study, not barrier-conditioned CVAE training or a DFT-validated diffusivity. Extensive ablations and optional extensions remain linked supplements to this report.

5.2 Independent checks and submission

Ricardo supplies physical/data/symmetry provenance, the energy-target and reference-accuracy interpretation, and final integration. Beihao independently reproduces a selected fit, reloads the checkpoint, verifies latent/graph/count behavior, and prioritizes a matched spatial no-latent control. Varad recomputes conditional response, novelty/yield, frozen mutation and cost accounting from saved attempt records. Prepared commands are not their completed contributions. The handoff specifies portable inputs, outputs and acceptance checks. Final integration is November 8; manual submission is due November 9, 2026 at 11:59 p.m. Pacific. Current course formatting instructions must be verified before submission.

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